SEMINAR III · WEEK 4

Fractions and Decimals: Different Ways to Show the Same Amount

This week we build on the number sense from Week 3. Fractions and decimals can represent the exact same quantity, just written two different ways. Learning to move between them, and to recognize when one form is more useful, is the goal of the week.

> **What quantity does this number represent?**

> **Would a fraction or decimal be more useful here?**

What Does a Fraction Mean?

Look at 3/4. The bottom number, 4, is the **denominator** — how many equal parts one whole has been divided into. The top number, 3, is the **numerator** — how many of those parts you have.

A fraction also means division: 3/4 also means 3 ÷ 4, which equals 0.75. This connection matters later this week, when we convert fractions to decimals.

Fractions are also points on a number line, not just pieces of a pie. One half is exactly halfway between 0 and 1. Three halves is greater than 1 — it sits halfway between 1 and 2. Fractions don't have to be less than 1.

Equivalent Fractions

Equivalent fractions represent the same quantity even though they look different — 1/2, 2/4, and 3/6 are all the same amount. Multiply the numerator and denominator by the same nonzero number and the value doesn't change: 1/2, multiplied top and bottom by 3, becomes 3/6. Same quantity, written differently.

Simplifying Fractions

Simplifying rewrites a fraction using smaller whole numbers, without changing its value. 12/18: both numbers divide by 6, giving 2/3. The way it's written changed. The quantity did not.

Comparing Fractions

Pick whichever method is fastest for the situation.

- **Same denominator:** compare how many pieces you have. 5/8 > 3/8.
- **Same numerator:** compare the size of the pieces. 3/4 > 3/8, because fourths are bigger than eighths.
- **Common denominator:** rewrite both fractions to share a denominator, then compare numerators.
- **Convert to decimal:** 3/4 = 0.75, 5/8 = 0.625, so 3/4 is greater.
- **Benchmarks:** compare each fraction to a familiar value like 1/2. 7/10 > 1/2, 3/8 < 1/2, so 7/10 is greater.

Adding and Subtracting Fractions

Fractions can only be added or subtracted directly when the denominators match. 1/5 + 2/5: pieces are already the same size, so just add numerators — 3/5. The denominator doesn't change, because the size of each piece never changed.

When denominators differ, like 1/2 + 1/4, first rewrite so they share a denominator: 1/2 becomes 2/4. Now add: 2/4 + 1/4 = 3/4.

> Imagine one person has half a pizza and another has a fourth of a pizza, from pizzas the same size. You can't just say “one piece plus one piece is two pieces” without first making sure the pieces are the same size.

Multiplying and Dividing Fractions

Multiplying fractions means finding “part of a part.” 1/2 × 3/4 means half of three fourths. Multiply straight across: 1×3=3, 2×4=8, so the answer is 3/8.

> Multiplication does **not** require a common denominator — that rule is only for addition and subtraction.

Dividing fractions asks a different question. 3/4 ÷ 1/2 asks: how many halves fit into 3/4? The answer is 1½. The shortcut: keep the first fraction, change division to multiplication, and use the reciprocal of the second fraction. It's fine to use the shortcut — but you should still understand what the division is actually asking.

Understanding Decimals

Each digit after the decimal point has a specific place value. In 4.583, the 5 is tenths, the 8 is hundredths, the 3 is thousandths — that 8 represents 0.08, not just “the number 8.”

Adding zeros to the right of a decimal doesn't change its value: 0.5 = 0.50 = 0.500. Useful when comparing decimals — rewrite with the same number of digits, then compare place by place. 0.6 vs 0.58 becomes 0.60 vs 0.58: 0.60 is greater.

> **Common mistake:** comparing decimals like whole numbers, thinking 0.58 > 0.6 because 58 looks bigger than 6. It isn't — place value, not digit count, decides.

Decimal Operations

Adding or subtracting: line up the decimal points so you're combining digits with the same place value.

> Multiplying, especially with a calculator: always estimate first. 4.9 × 3.1 should land close to 5×3=15. If your calculator shows 151.9, something went wrong — probably a decimal point in the wrong place.

Dividing: think about what the division actually represents. If $12.60 is split evenly three ways, each person gets $4.20.

Fractions and Decimals Are Connected

Since a fraction means division, convert a fraction to a decimal by dividing: 3/4 = 3÷4 = 0.75. Going the other direction, a decimal becomes a fraction using its place value: 0.6 means six tenths, 6/10, which simplifies to 3/5.

These benchmark conversions come up constantly and are worth memorizing:

1/2 = 0.5 | 1/4 = 0.25 | 3/4 = 0.75 | 1/5 = 0.2 | 2/5 = 0.4 | 3/5 = 0.6 | 4/5 = 0.8 | 1/10 = 0.1

These benchmarks matter again next week, when we introduce percent.

Which Form Should I Use?

Money is almost always a decimal: $2.75, not $2¾. Recipes usually use fractions: 3/4 cup, not 0.75 cup. When comparing two values, decimals can make it easier: converting 7/8 to 0.875 makes it immediately clear that 7/8 > 0.82.

Does My Answer Make Sense?

Check	Ask
Operation	Did I use the correct operation?
Size	Should this answer be greater than 1, or less than 1?
Sign	Should the answer be positive, negative, or zero?
Unit	Did I answer in the correct unit?
Context	Does this answer make sense here?

> For fractions and decimals, **Size** deserves extra attention: multiplying two numbers less than 1 should make the result *smaller*. If a student multiplies 1/2 × 1/3 and gets 5/6, that should immediately look wrong.

Error Detective

Knowledge — 1/2 + 1/3 = 2/5?

The student added numerators and denominators directly. The rule for adding fractions with unlike denominators wasn't correctly known or applied. Fix: find a common denominator first.

Execution — 2.4 × 3 = 72?

Estimating first, 2×3 ≈ 6, so 72 is clearly unreasonable — likely a decimal-placement mistake.

Strategy — comparing 1/2 and 0.45

A student spends several minutes finding a common denominator instead of just recognizing 1/2 = 0.5. A faster, more useful approach was available.

Also watch for: a student saying 1/8 is greater than 1/4, because 8 > 4. This ignores what the denominator means — more pieces means smaller pieces. 1/8 < 1/4.

Quick Reference

Numerator: how many parts you have.

Denominator: how many equal parts make one whole.

Adding/subtracting fractions: requires a common denominator.

Multiplying fractions: multiply straight across, no common denominator needed.

Dividing fractions: keep the first fraction, change to multiplication, use the reciprocal of the second.

Common benchmarks: 1/2=0.5, 1/4=0.25, 3/4=0.75, 1/5=0.2, 1/10=0.1.

Check your answer: Operation, Size, Sign, Unit, Context.

Keep this page. You'll use it all year.